Supplemental Material: Prompt Language Doesn't Matter, but Keep the Original Text: Large Language Models and Sentiment Analysis in French News

Abstract

Supplemental tables and figures for the article.

Keywords: supplemental material

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Table S1
Search Terms Used to Select Articles in the Eureka Database

Group	Search terms
French terms	*logiciel libre*, *logiciels libres*, *logiciel open source*, *logiciels open source*, *code source ouvert*, *code source libre*
English and borrowed terms	*open source*, *open-source*, *free software*, *software libre*
Proper names	*Free Software Foundation*, *Richard Stallman*

Table S2
Collected Articles by Source Media

Source	Articles	First article	Last article
Libération	769	1995-06-23	2024-11-04
Monde, Le	666	2001-01-09	2024-12-26
Figaro, Le	402	1998-10-21	2024-12-04
Devoir, Le	318	1998-07-20	2024-10-26
Soleil, Le (Québec, QC)	179	1998-11-08	2023-11-18
Presse, La	152	1998-03-26	2017-10-14
Journal de Montréal, Le	65	2006-12-20	2023-04-24
Journal de Québec, Le	45	2007-03-24	2024-10-19
Tribune, La (Sherbrooke, QC)	43	2000-03-29	2022-12-03
Droit, Le (Ottawa, ON)	17	1998-11-10	2014-07-17
Nouvelliste, Le (Trois-Rivières, QC)	13	2000-01-13	2022-03-05
Quotidien, Le (Saguenay, QC)	10	2008-11-01	2019-04-10
Voix de l'Est, La (Granby, QC)	4	2010-09-17	2015-02-14

Table S3

Wording of the English and French Prompts on French Text, by Component. Both Were Preceded by the Same English System Prompt: "You are a helpful assistant that analyzes the sentiment of text. You provide accurate and consistent sentiment ratings according to the specified scale." The English Prompt on the English Translation Differs Only in Referring to English Instead of French (Task and Instruction 1).

Component	English prompt	French prompt
Task	Please analyze the sentiment of the following French text and provide a single numerical rating according to this scale:	Veuillez analyser le sentiment du texte français suivant et fournir une évaluation numérique unique selon cette échelle :
Scale, -1.0	Strong negative sentiment - highly critical, hostile, or pessimistic content	Sentiment négatif fort - contenu très critique, hostile ou pessimiste
Scale, -0.5	Moderate negative sentiment - somewhat negative, disapproving, or concerned content	Sentiment négatif modéré - contenu plutôt négatif, désapprobateur ou préoccupant
Scale, 0.0	Neutral sentiment - factual, balanced, or neither positive nor negative content	Sentiment neutre - contenu factuel, équilibré, ou ni positif ni négatif
Scale, 0.5	Moderate positive sentiment - somewhat positive, approving, or optimistic content	Sentiment positif modéré - contenu plutôt positif, approbateur ou optimiste
Scale, 1.0	Strong positive sentiment - highly supportive, enthusiastic, or optimistic content	Sentiment positif fort - contenu très favorable, enthousiaste ou optimiste
Intermediate values	You may also use values between these points for intermediate sentiment levels (e.g., -0.7, 0.3).	Vous pouvez également utiliser des valeurs intermédiaires entre ces points pour des niveaux de sentiment intermédiaires (par exemple, -0.7, 0.3).
Instruction 1	First, carefully read and understand the text, considering cultural and linguistic nuances in French.	D'abord, lisez attentivement et comprenez le texte, en tenant compte des nuances culturelles et linguistiques en français.
Instruction 2	Analyze the emotional tone, word choice, and overall message.	Analysez le ton émotionnel, le choix des mots et le message global.
Instruction 3	Respond ONLY with a single numerical value between -1.0 and 1.0 that best represents the sentiment.	Répondez UNIQUEMENT avec une valeur numérique unique entre -1.0 et 1.0 qui représente le mieux le sentiment.
Instruction 4	Do not include ANY explanations, analysis, or additional text in your response.	N'incluez AUCUNE explication, analyse ou texte supplémentaire dans votre réponse.
Input	Here is the text to analyze: [sentence]	Voici le texte à analyser : [sentence]

Table S4

Correlation With Human Judgments (With 95% Confidence Interval), Mean Absolute Error, and Number of Sentences Scored, by Model and Condition

Model	Weights	Condition	r	95% CI	MAE	n
GPT-5.6 Luna	Closed	FR→FR	0.810	[0.76, 0.85]	0.168	200
GPT-5.6 Luna	Closed	EN→FR	0.802	[0.75, 0.85]	0.165	200
GPT-5.6 Luna	Closed	EN→EN	0.728	[0.66, 0.79]	0.176	200
Jev 1.13	Closed	FR→FR	0.772	[0.71, 0.82]	0.163	200
Jev 1.13	Closed	EN→FR	0.772	[0.71, 0.82]	0.169	200
Jev 1.13	Closed	EN→EN	0.752	[0.68, 0.81]	0.176	200
Gemini 3.5 Flash	Closed	FR→FR	0.760	[0.69, 0.81]	0.170	200
Gemini 3.5 Flash	Closed	EN→FR	0.773	[0.71, 0.82]	0.162	200
Gemini 3.5 Flash	Closed	EN→EN	0.760	[0.69, 0.81]	0.165	200
Claude Haiku 4.5	Closed	FR→FR	0.785	[0.73, 0.83]	0.182	200
Claude Haiku 4.5	Closed	EN→FR	0.780	[0.72, 0.83]	0.185	200
Claude Haiku 4.5	Closed	EN→EN	0.705	[0.63, 0.77]	0.182	200
Qwen3 235B-A22B	Open	FR→FR	0.774	[0.71, 0.82]	0.226	200
Qwen3 235B-A22B	Open	EN→FR	0.753	[0.69, 0.81]	0.232	200
Qwen3 235B-A22B	Open	EN→EN	0.680	[0.60, 0.75]	0.229	200
GPT-OSS 120B	Open	FR→FR	0.727	[0.65, 0.79]	0.205	200
GPT-OSS 120B	Open	EN→FR	0.743	[0.67, 0.80]	0.182	200
GPT-OSS 120B	Open	EN→EN	0.729	[0.66, 0.79]	0.172	200
GPT-OSS 20B	Open	FR→FR	0.733	[0.66, 0.79]	0.171	200
GPT-OSS 20B	Open	EN→FR	0.718	[0.64, 0.78]	0.191	200
GPT-OSS 20B	Open	EN→EN	0.695	[0.62, 0.76]	0.176	200
Mistral Small 3.2	Open	FR→FR	0.735	[0.66, 0.79]	0.263	200
Mistral Small 3.2	Open	EN→FR	0.728	[0.66, 0.79]	0.294	200
Mistral Small 3.2	Open	EN→EN	0.669	[0.58, 0.74]	0.272	200
DeepSeek V3.2	Open	FR→FR	0.686	[0.61, 0.75]	0.203	200
DeepSeek V3.2	Open	EN→FR	0.743	[0.67, 0.80]	0.176	200
DeepSeek V3.2	Open	EN→EN	0.683	[0.60, 0.75]	0.178	200
Mistral Large 4	Closed	FR→FR	0.702	[0.62, 0.77]	0.184	200
Mistral Large 4	Closed	EN→FR	0.713	[0.64, 0.78]	0.176	200
Mistral Large 4	Closed	EN→EN	0.692	[0.61, 0.76]	0.169	200
Qwen3 32B	Open	FR→FR	0.679	[0.60, 0.75]	0.175	200
Qwen3 32B	Open	EN→FR	0.695	[0.62, 0.76]	0.170	200
Qwen3 32B	Open	EN→EN	0.644	[0.56, 0.72]	0.187	200
Llama 4 Scout	Open	FR→FR	0.644	[0.55, 0.72]	0.275	200
Llama 4 Scout	Open	EN→FR	0.661	[0.58, 0.73]	0.286	200
Llama 4 Scout	Open	EN→EN	0.618	[0.52, 0.70]	0.299	200
Llama 3.1 8B	Open	FR→FR	0.612	[0.52, 0.69]	0.273	199
Llama 3.1 8B	Open	EN→FR	0.604	[0.51, 0.68]	0.268	200
Llama 3.1 8B	Open	EN→EN	0.575	[0.47, 0.66]	0.254	200
Lexicoder (EN)	Dictionary	EN	0.548	[0.44, 0.64]	0.203	200
Lexicoder (FR)	Dictionary	FR	0.357	[0.23, 0.47]	0.210	200
Llama 3.2 3B	Open	FR→FR	0.476	[0.36, 0.58]	0.262	200
Llama 3.2 3B	Open	EN→FR	0.132	[-0.01, 0.27]	0.377	200
Llama 3.2 3B	Open	EN→EN	0.302	[0.17, 0.42]	0.305	200
Llama 3.2 1B	Open	FR→FR	0.100	[-0.16, 0.35]	0.551	58
Llama 3.2 1B	Open	EN→FR	0.003	[-0.23, 0.23]	0.604	74
Llama 3.2 1B	Open	EN→EN	-0.199	[-0.42, 0.05]	0.526	66

Table S5

Cost of Scoring 1,000 Sentences at List Prices (Upper Bound; Cached Prompt Tokens Counted at the Full Input Rate; Input Tokens and Cost Not Reported for Llama 3.2 1B, Whose Endpoint Returned a Constant Token Count)

Model	Provider	Weights	API calls per score	Input tokens	Output tokens	USD per 1,000 sentences
GPT-OSS 20B	OpenRouter	Open	1.02	657,496	87,892	0.018
Llama 3.2 3B	OpenRouter	Open	1.00	595,142	7,320	0.018
Jev 1.13	TypeSafe	Closed	1.00	815,199	32,400	0.019
GPT-OSS 120B	OpenRouter	Open	1.00	667,425	95,660	0.020
Qwen3 32B	OpenRouter	Open	1.00	592,086	5,696	0.027
Mistral Small 3.2	OpenRouter	Open	1.00	553,761	7,405	0.029
Qwen3 235B-A22B	OpenRouter	Open	1.00	584,886	7,752	0.030
Llama 4 Scout	OpenRouter	Open	1.00	564,947	8,467	0.059
DeepSeek V3.2	OpenRouter	Open	1.00	559,594	9,340	0.067
Llama 3.1 8B	OpenRouter	Open	1.01	593,398	8,062	0.074
GPT-5.6 Luna	OpenAI	Closed	1.00	552,225	51,819	0.096
Mistral Large 4	OpenRouter	Closed	1.00	534,990	7,603	0.211
Claude Haiku 4.5	Anthropic	Closed	1.00	648,705	12,600	0.395
Gemini 3.5 Flash	Google	Closed	1.04	572,330	577,590	3.365
Llama 3.2 1B	OpenRouter	Open	22.68	–	168,789	–

Table S6
F1 Scores for 3-Category Classification

Model	Negative	Neutral	Positive	Weighted F1
Claude Haiku 4.5 (FR→FR)	0.762	0.353	0.834	0.770
GPT-5.6 Luna (FR→FR)	0.748	0.400	0.811	0.756
Jev 1.13 (EN→FR)	0.691	0.444	0.821	0.748
Jev 1.13 (FR→FR)	0.695	0.357	0.814	0.738
GPT-5.6 Luna (EN→FR)	0.730	0.345	0.787	0.732
Claude Haiku 4.5 (EN→FR)	0.690	0.290	0.789	0.715
Jev 1.13 (EN→EN)	0.711	0.200	0.787	0.713
Mistral Small 3.2 (FR→FR)	0.578	0.250	0.832	0.703
DeepSeek V3.2 (FR→FR)	0.672	0.300	0.766	0.697
Mistral Small 3.2 (EN→FR)	0.545	0.320	0.829	0.697
Qwen3 235B-A22B (FR→FR)	0.752	0.260	0.727	0.695
GPT-5.6 Luna (EN→EN)	0.688	0.349	0.746	0.694
DeepSeek V3.2 (EN→FR)	0.696	0.250	0.748	0.689
GPT-OSS 120B (FR→FR)	0.721	0.301	0.722	0.686
Llama 4 Scout (FR→FR)	0.643	0.310	0.748	0.678
Qwen3 235B-A22B (EN→FR)	0.719	0.243	0.717	0.677
Mistral Small 3.2 (EN→EN)	0.488	0.320	0.824	0.675
Llama 4 Scout (EN→FR)	0.667	0.277	0.733	0.673
GPT-OSS 120B (EN→FR)	0.717	0.312	0.700	0.672
Claude Haiku 4.5 (EN→EN)	0.678	0.247	0.725	0.670
GPT-OSS 20B (EN→FR)	0.700	0.282	0.687	0.657
GPT-OSS 20B (FR→FR)	0.691	0.271	0.691	0.655
Llama 4 Scout (EN→EN)	0.647	0.233	0.696	0.641
DeepSeek V3.2 (EN→EN)	0.672	0.246	0.657	0.626
Mistral Large 4 (EN→EN)	0.604	0.242	0.687	0.623
Gemini 3.5 Flash (EN→EN)	0.667	0.244	0.653	0.622
Llama 3.1 8B (EN→FR)	0.652	0.269	0.656	0.622
Mistral Large 4 (FR→FR)	0.644	0.250	0.660	0.620
Llama 3.1 8B (EN→EN)	0.627	0.279	0.663	0.619
Gemini 3.5 Flash (FR→FR)	0.649	0.253	0.649	0.615
Mistral Large 4 (EN→FR)	0.648	0.229	0.642	0.609
Gemini 3.5 Flash (EN→FR)	0.661	0.231	0.630	0.606
Llama 3.2 3B (FR→FR)	0.465	0.182	0.733	0.602
Llama 3.1 8B (FR→FR)	0.615	0.163	0.653	0.602
GPT-OSS 20B (EN→EN)	0.614	0.227	0.646	0.600
Qwen3 235B-A22B (EN→EN)	0.698	0.220	0.601	0.599
Lexicoder (EN)	0.574	0.243	0.646	0.589
GPT-OSS 120B (EN→EN)	0.678	0.231	0.562	0.570
Qwen3 32B (EN→FR)	0.554	0.179	0.631	0.568
Qwen3 32B (FR→FR)	0.515	0.207	0.620	0.552
Lexicoder (FR)	0.474	0.279	0.620	0.545
Qwen3 32B (EN→EN)	0.559	0.202	0.529	0.511
Llama 3.2 3B (EN→EN)	0.167	0.245	0.717	0.503
Llama 3.2 3B (EN→FR)	0.000	0.200	0.745	0.464
Llama 3.2 1B (FR→FR)	0.000	0.194	0.638	0.423
Llama 3.2 1B (EN→FR)	0.000	0.057	0.558	0.321

Table S7

F1 Scores for 7-Category Classification

Model	Very neg.	Negative	Somewhat neg.	Neutral	Somewhat pos.	Positive	Very pos.	Weighted F1
Claude Haiku 4.5 (FR→FR)	0.400	0.680	0.455	0.353	0.537	0.612	0.000	0.532
Jev 1.13 (EN→FR)	0.200	0.586	0.282	0.444	0.634	0.548	0.250	0.527
Jev 1.13 (FR→FR)	0.222	0.621	0.324	0.357	0.613	0.540	0.250	0.521
Jev 1.13 (EN→EN)	0.200	0.525	0.385	0.200	0.621	0.566	0.286	0.515
GPT-5.6 Luna (FR→FR)	0.286	0.500	0.393	0.400	0.549	0.462	0.133	0.483
GPT-5.6 Luna (EN→EN)	0.364	0.500	0.338	0.349	0.521	0.542	0.250	0.471
GPT-5.6 Luna (EN→FR)	0.333	0.500	0.258	0.345	0.575	0.407	0.125	0.456
Claude Haiku 4.5 (EN→EN)	0.286	0.642	0.400	0.247	0.419	0.568	0.000	0.449
DeepSeek V3.2 (EN→EN)	0.333	0.533	0.419	0.246	0.409	0.588	0.667	0.442
GPT-OSS 120B (EN→FR)	0.333	0.566	0.400	0.312	0.426	0.485	0.133	0.436
Mistral Large 4 (EN→EN)	0.333	0.651	0.316	0.242	0.454	0.408	0.000	0.423
DeepSeek V3.2 (EN→FR)	0.444	0.667	0.429	0.250	0.358	0.493	0.000	0.420
DeepSeek V3.2 (FR→FR)	0.333	0.368	0.469	0.300	0.394	0.527	0.167	0.417
Mistral Large 4 (EN→FR)	0.167	0.600	0.357	0.229	0.370	0.590	0.286	0.417
Claude Haiku 4.5 (EN→FR)	0.500	0.680	0.364	0.290	0.356	0.511	0.200	0.417
Mistral Large 4 (FR→FR)	0.154	0.488	0.344	0.250	0.406	0.561	0.222	0.413
Mistral Small 3.2 (EN→EN)	0.333	0.345	0.222	0.320	0.525	0.364	0.091	0.399
Gemini 3.5 Flash (EN→EN)	0.222	0.604	0.310	0.244	0.333	0.571	0.286	0.392
GPT-OSS 20B (EN→FR)	0.235	0.431	0.269	0.282	0.379	0.535	0.000	0.379
Mistral Small 3.2 (FR→FR)	0.500	0.400	0.170	0.250	0.482	0.380	0.125	0.376
GPT-OSS 120B (EN→EN)	0.444	0.692	0.281	0.231	0.296	0.536	0.286	0.376
GPT-OSS 120B (FR→FR)	0.308	0.545	0.333	0.301	0.328	0.447	0.154	0.371
GPT-OSS 20B (FR→FR)	0.364	0.600	0.286	0.271	0.333	0.448	0.286	0.370
GPT-OSS 20B (EN→EN)	0.400	0.528	0.235	0.227	0.346	0.508	0.000	0.363
Gemini 3.5 Flash (FR→FR)	0.400	0.600	0.222	0.253	0.283	0.471	0.200	0.339
Qwen3 32B (EN→FR)	0.222	0.512	0.163	0.179	0.293	0.615	0.000	0.337
Qwen3 235B-A22B (FR→FR)	0.190	0.407	0.320	0.260	0.327	0.369	0.087	0.334
Gemini 3.5 Flash (EN→FR)	0.571	0.679	0.245	0.231	0.212	0.514	0.200	0.329
Qwen3 32B (FR→FR)	0.333	0.558	0.125	0.207	0.243	0.632	0.000	0.320
Qwen3 235B-A22B (EN→FR)	0.182	0.321	0.240	0.243	0.359	0.339	0.091	0.316
Mistral Small 3.2 (EN→FR)	0.364	0.312	0.133	0.320	0.390	0.292	0.062	0.309
Qwen3 235B-A22B (EN→EN)	0.182	0.275	0.302	0.220	0.286	0.408	0.133	0.300
Llama 3.1 8B (EN→FR)	0.111	0.120	0.192	0.269	0.327	0.507	0.200	0.299
Llama 4 Scout (FR→FR)	0.125	0.167	0.182	0.310	0.333	0.356	0.100	0.284
Lexicoder (FR)	0.000	0.000	0.253	0.279	0.463	0.057	0.000	0.279
Lexicoder (EN)	0.000	0.000	0.267	0.243	0.481	0.000	0.000	0.277
Llama 3.1 8B (FR→FR)	0.143	0.197	0.179	0.163	0.245	0.543	0.000	0.268
Llama 3.2 3B (FR→FR)	0.000	0.364	0.200	0.182	0.246	0.395	0.000	0.267
Llama 3.1 8B (EN→EN)	0.138	0.222	0.136	0.279	0.252	0.479	0.143	0.265
Qwen3 32B (EN→EN)	0.222	0.364	0.172	0.202	0.173	0.567	0.333	0.264
Llama 4 Scout (EN→FR)	0.154	0.261	0.143	0.277	0.270	0.329	0.080	0.254
Llama 4 Scout (EN→EN)	0.105	0.280	0.125	0.233	0.241	0.434	0.154	0.253
Llama 3.2 3B (EN→EN)	0.000	0.138	0.050	0.245	0.256	0.375	0.000	0.219
Llama 3.2 3B (EN→FR)	0.000	0.000	0.000	0.200	0.239	0.317	0.000	0.172
Llama 3.2 1B (EN→EN)	0.000	0.000	0.000	0.146	0.000	0.211	0.000	0.053
Llama 3.2 1B (FR→FR)	0.000	0.000	0.000	0.194	0.000	0.000	0.000	0.017
Llama 3.2 1B (EN→FR)	0.000	0.000	0.000	0.057	0.000	0.000	0.000	0.004

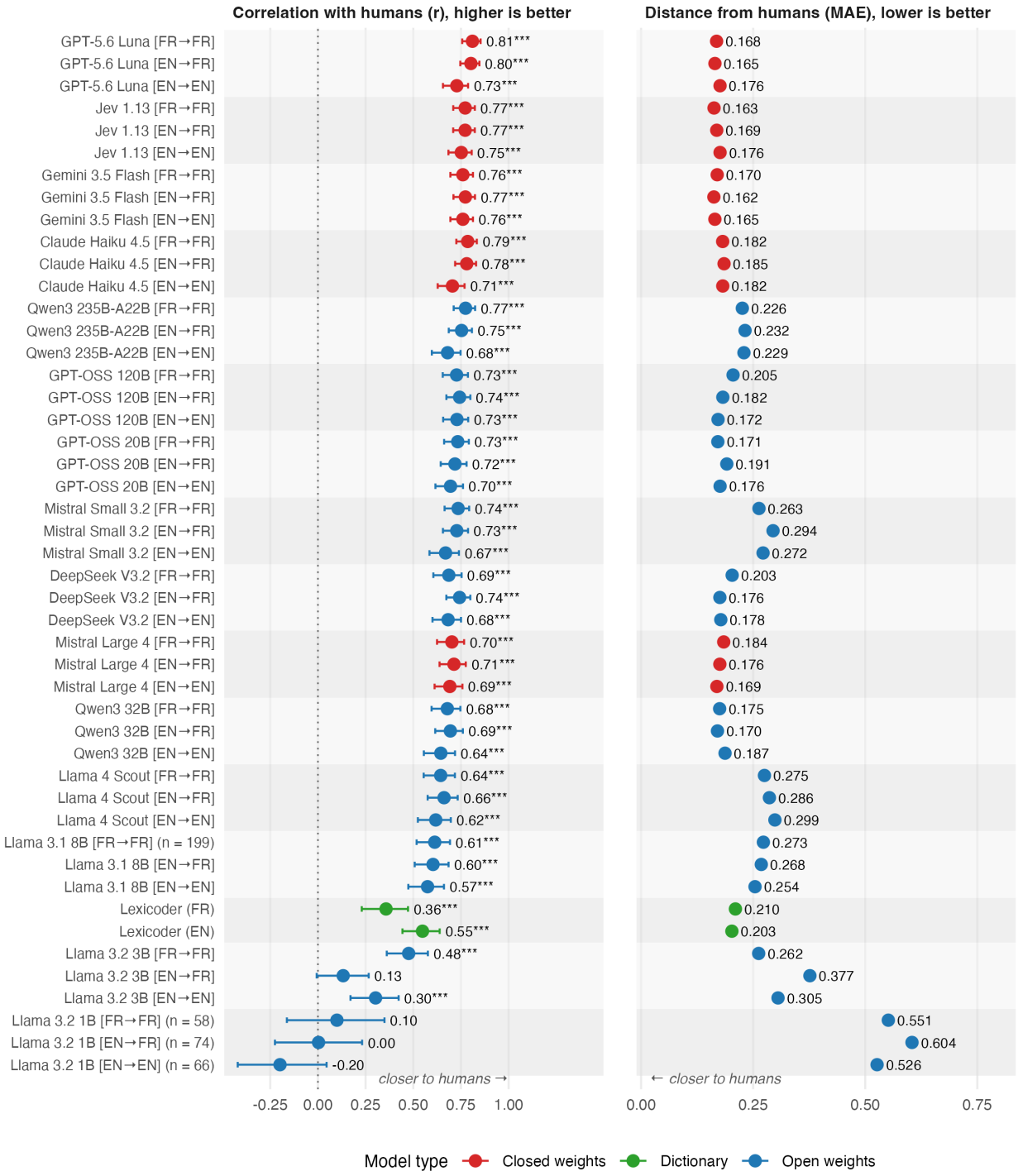
Table S8

Mean Input Tokens per API Call by Language Condition, and Tokens per Word by Text Language (Valid Calls Only; Llama 3.2 1B Excluded Because Its Endpoint Reported a Constant Token Count on Every Call; Jev's Input Is a Structured Question Rather Than the Text Prompt, So Its Counts Are Not Comparable)

Model	EN→EN	EN→FR	FR→FR	French-prompt overhead	Tokens per French word	Tokens per English word
Mistral Large 4	284.5	289.1	318.1	+29 (+10.0%)	1.38	1.22
Jev 1.13	431.0	440.3	487.3	+47 (+10.7%)	1.62	1.22
GPT-OSS 120B	352.0	357.7	402.7	+45 (+12.6%)	1.42	1.19
GPT-OSS 20B	339.0	344.7	389.7	+45 (+13.1%)	1.42	1.19
Llama 4 Scout	294.1	299.9	343.9	+44 (+14.7%)	1.41	1.19
Mistral Small 3.2	288.6	294.6	339.6	+45 (+15.3%)	1.45	1.22
GPT-5.6 Luna	288.0	293.7	338.7	+45 (+15.3%)	1.42	1.19
Gemini 3.5 Flash	285.0	292.1	337.1	+45 (+15.4%)	1.46	1.17
Qwen3 32B	296.5	307.7	382.7	+75 (+24.4%)	1.69	1.21
DeepSeek V3.2	280.1	290.3	361.3	+71 (+24.5%)	1.61	1.19
Qwen3 235B-A22B	292.5	303.7	378.7	+75 (+24.7%)	1.69	1.21
Llama 3.2 3B	295.2	306.8	388.8	+82 (+26.7%)	1.69	1.20
Llama 3.1 8B	295.2	306.8	389.0	+82 (+26.8%)	1.69	1.20
Claude Haiku 4.5	313.8	328.2	439.2	+111 (+33.8%)	1.87	1.25

Figure S1

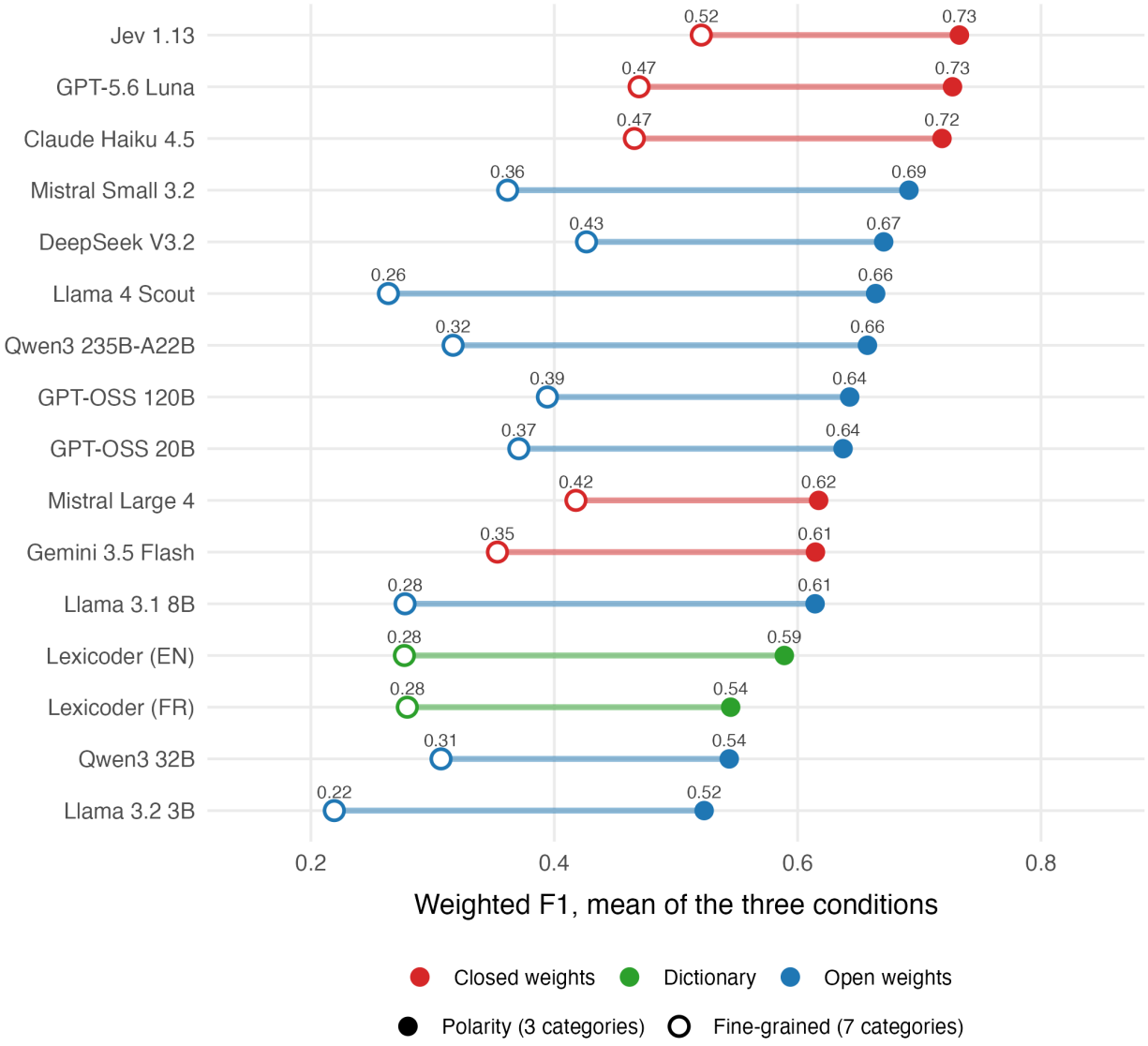
Correlation and Mean Absolute Error Between Model Scores and Human Judgments, by Model and Condition



Note. One row per model and condition, sorted by each model's mean correlation and colored by model type. Left panel: Pearson correlation between the model's scores and the mean of three human coders, with 95% confidence intervals. Right panel: mean absolute error, the average distance between the model's score and the human score on the −1 to 1 scale, so that 0.17 means the model is typically 0.17 points away from the human rating. Within each model, the conditions are FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text). $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Figure S2

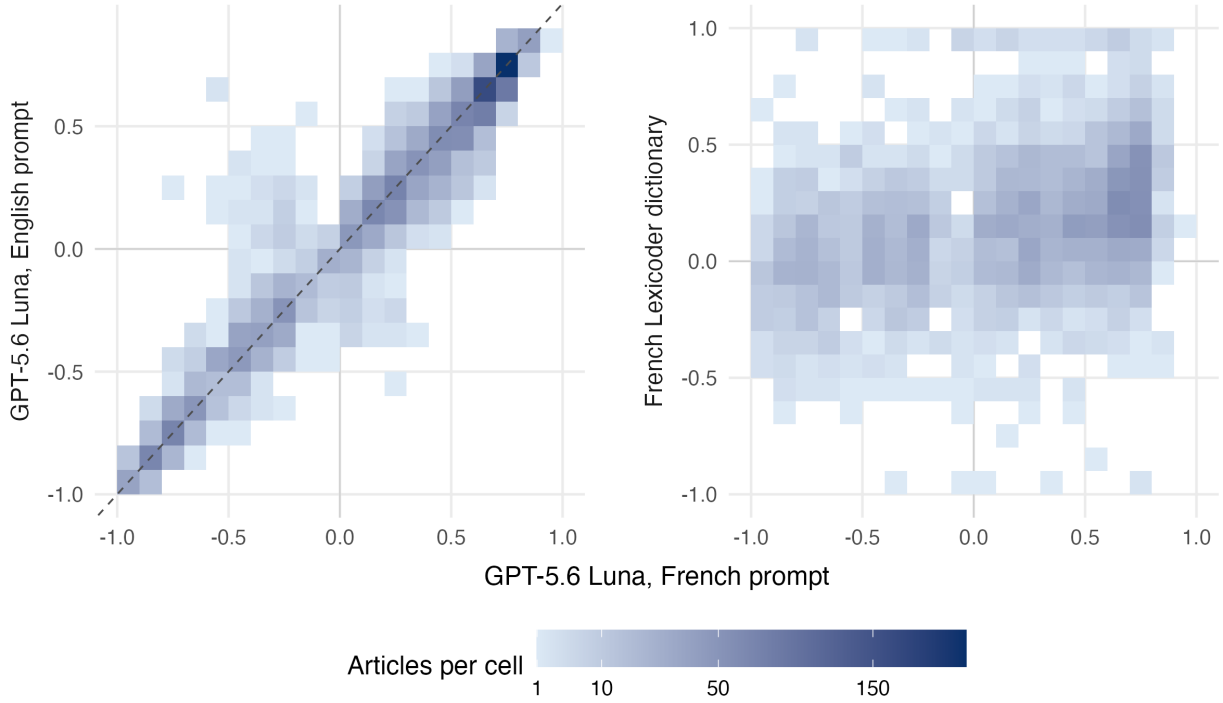
Weighted F1 on Polarity and on Fine-Grained Classification, by Model



Note. One row per model, from the highest polarity F1 to the lowest, colored by model type. Filled circles are the weighted F1 on three categories, hollow circles on seven, each averaged over the three conditions; the segment is the loss from grouping sentiment into seven categories instead of three. Llama 3.2 1B is omitted.

Figure S3

Agreement Between Scoring Methods on the Full Corpus, by Article



Note. Each cell covers a 0.1 by 0.1 range of scores and is shaded by the number of articles it contains (n = 2,671). The horizontal axis is the GPT-5.6 Luna score under the French prompt. Left: the same model's score under the English prompt, with the identity line dashed (r = 0.97; the two prompts give the same sign on 94% of articles). Right: the French Lexicoder score on the original text (r = 0.36; the dictionary and the model give the same sign on 66% of articles and opposite signs on 28%, the remainder being articles that one of the two scores exactly zero). All scores run from −1 (very negative) to 1 (very positive).